
Medical Insurance Claim Fraud Detection

Approach 1: Traditional Machine Learning

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Abstract

This report presents a reproducible traditional machine-learning baseline for medical insurance claim fraud screening in an Indian healthcare context. The repository-provided workbook contains 4,500 claim rows, of which 270 are labeled fraud. The pipeline validates the schema, profiles missingness, creates screening-time financial and temporal features, excludes identifiers and likely post-decision fields, and fits imputation, one-hot encoding, and scaling only on the training partition. A stratified 70/15/15 split supports validation-based model selection and a locked test evaluation. A zoo of linear, probabilistic, tree, boosting, margin, instance, neural-baseline, and ensemble classifiers is compared using precision, recall, F1, F2, ROC-AUC, PR-AUC, MCC, calibration, latency, and artifact size. The winner is selected using validation F2 with PR-AUC tie-breaking and then refit once before test evaluation. The output is a triage recommendation, not an automatic denial. Results are limited by the 4,500-row supplied snapshot, missing policy and document fields, and uncertain provenance; these limitations motivate the deep-learning and agentic extensions.

Keywords: fraud detection; health insurance; classification; imbalanced learning; explainable AI; India

1. Introduction

Medical insurance fraud includes exaggerated amounts, duplicate submissions, services not rendered, misrepresented clinical details, and collusion. Manual review is expensive and cannot inspect every record with equal depth. A screening model can prioritize records for human investigation while preserving due process.

The project asks whether a structured claim record can support a defensible fraud probability and binary fraud flag. The primary audience is an academic reviewer who must be able to trace each number to code, data, configuration, and artifact. The system therefore treats reproducibility and interpretability as first-class requirements.

The project is not an autonomous adjudication system. A model score is one signal among policy terms, medical documentation, provider records, and investigator judgment. This framing is essential where false positives can harm genuine policyholders and false negatives can create financial loss.

2. Indian insurance context and responsible use

Indian health insurance includes individual, family-floater, senior-citizen, employer group, and public programme contexts. Real claims vary by city tier, hospital network, treatment type, documentation quality, and policy wording. A responsible project should represent this diversity rather than assume a single foreign claims process.

The supplied workbook does not contain verified Indian state, policy, hospital-tier, sum-insured, or document fields. We do not invent them. Instead, the report records Indian context as a design requirement and identifies the missing variables as future data-collection priorities. This is more defensible than relabeling fictional locations as Indian geography.

The model is intended to flag and explain, not accuse. Any real deployment would require privacy controls, IRDAI-aligned claim communication, human review, appeal mechanisms, and periodic fairness and calibration audits.

3. Dataset and data quality

The workbook has 4,500 rows and 19 source columns. The target is ClaimLegitimacy with Fraud as the positive class and Legitimate as the negative class. The fraud prevalence is 6.0%. There are 0 exact duplicate rows and 0 duplicate ClaimID values in the validated snapshot; missing cells total 0.

ClaimAmount, PatientIncome, PatientAge, Cluster, date, demographic categories, claim type, submission method, specialty, diagnosis code, procedure code, and identifiers are present. Diagnosis and procedure codes are nearly unique, while provider location is high-cardinality. IDs are not meaningful reusable evidence for unseen claims.

The dataset card describes the workbook as repository-provided academic data with provenance and redistribution terms requiring confirmation. Reported metrics should not be generalized to all insurers or all Indian policyholders.

Partition	Rows	Fraud	Fraud rate
Train	3150	189	6.0%
Validation	675	41	6.1%
Test	675	40	5.9%

4. Exploratory data analysis

EDA is executed before modeling. It includes class balance, missingness, numerical distributions, class-conditioned box plots, rank correlations, categorical counts and rates, financial relationships, temporal volume, and a claim-type/specialty fraud-rate heatmap. Figures are generated programmatically at high resolution and indexed in documentation/figure_index.md.

The supplied fraud rows show higher typical claim values and lower reported income, while age and most low-cardinality categories overlap substantially. Cluster is strongly concentrated in fraud rows; this may be a useful supplied signal or a synthetic shortcut. The correct conclusion is to retain it with an explicit audit note and test stability on a future external dataset.

EDA findings guide transformations but do not establish causation. Sparse category rates are shown with a stability caveat. The visual identity uses teal for legitimate and orange for fraud across all deliverables.

5. Preprocessing and feature engineering

The pipeline first creates screening-time features: log amounts, claim-to-income ratio, scaled amount-income difference, amount per age, calendar components, weekend flag, cyclic month encoding, age bands, and normalized low-cardinality categories. The target and identifier fields are kept outside the feature matrix.

Median numeric imputation and standard scaling are fitted on training rows. Categorical imputation and one-hot encoding use an unknown-safe encoder. This common transformer produces the same feature order for every model and is serialized for inference. Validation and test matrices are transformed, never refit.

Raw ClaimStatus is excluded because it may be updated after adjudication. Raw codes and location are excluded because their cardinality is nearly the row count. These choices trade some available signal for better generalization and lower memorization risk.

6. Algorithms and tuning

The benchmark includes a majority baseline, logistic L1/L2, decision tree, random forest, extra trees, gradient boosting, histogram gradient boosting, AdaBoost, RBF SVM, KNN, Gaussian and Bernoulli naive Bayes, LDA, QDA, MLP, calibrated ridge, passive-aggressive, soft voting, and stacking. Each model is created from a declarative registry with a stable key and search space.

Search uses stratified three-fold cross-validation on training rows and average precision as the tuning score. Search budgets are deliberately small and recorded. Models that fail remain in the audit trail. Training seconds, prediction latency, and serialized artifact size support an efficiency discussion.

External XGBoost, LightGBM, and CatBoost are documented as optional future adapters; the core run uses pinned scikit-learn dependencies so the repository remains runnable on a laptop without GPU support.

7. Evaluation and thresholding

The metric suite includes accuracy, balanced accuracy, fraud precision, fraud recall, F1, F2, ROC-AUC, PR-AUC, MCC, Cohen's kappa, Brier score, log loss, specificity, and the confusion matrix. F2 is used for operating-point selection because recall is operationally important, while PR-AUC summarizes ranking quality under imbalance.

The validation policy selected Soft voting ensemble with validation F2 1.0000, validation PR-AUC 1.0000, and threshold 0.4900. The exact leaderboard and test values are in evaluation/leaderboard.csv and evaluation/evaluation.md.

After selection, the winner is refit on training plus validation rows and evaluated on the held-out test partition. Bootstrap intervals, calibration, and fairness tables are generated as supporting evidence. No result is hand-entered into this report.

Model	Val F2	Val PR-AUC	Test F2	Test PR-AUC
Random forest	1.0000	1.0000	0.9950	0.9994
Gradient boosting	1.0000	1.0000	0.9799	0.9994
Histogram gradient boostin	1.0000	1.0000	0.9799	0.9994
AdaBoost	1.0000	1.0000	0.9799	0.9885
Soft voting ensemble	1.0000	1.0000	0.9799	1.0000
Stacking ensemble	1.0000	1.0000	0.9950	1.0000
Support vector machine (RB	0.9804	0.9820	0.9406	0.9800
Decision tree	0.9804	0.9771	0.9799	0.9765

8. Explainability and fairness

Permutation importance is computed on validation data for the selected model. The result is a global association ranking. Local explanation templates provide the score band, strongest signals, caveats, and next steps in neutral language. They avoid claiming that a claimant intended or committed fraud.

Fairness slices report recall, false-positive rate, precision, accuracy, and predicted fraud rate across gender, age bands, claim type, and employment status. Small cells are marked unstable. A disparity trigger initiates investigation rather than a claim of bias because the data is a supplied academic snapshot.

For regulated use, explanations must be coupled with verified policy clauses, document evidence, an appeal route, and human adjudication. Calibration and explanation quality should be monitored across groups, not only overall.

9. Discussion

The traditional approach is attractive because it is fast, comparatively interpretable, and well suited to structured tabular data. It provides a useful baseline for later neural and agentic approaches. Its weaknesses are equally important: it cannot read documents, it relies on manually engineered interactions, and its probability quality can degrade under temporal or geographic shift.

The workbook's high-cardinality and cluster structure mean the strongest observed performance may reflect synthetic construction. A future evaluation should use a time split, an external provider split, and a license-cleared Indian claims source. Results should be compared with the same target semantics and locked test cases.

The project contribution is therefore a transparent implementation and artifact chain rather than a claim of universal superiority. The model suite, leakage register, threshold memo, fairness audit, and generated deliverables make the baseline useful for learning and comparison.

10. Conclusion and future work

This report delivers a runnable, reproducible traditional ML pipeline for fraud screening with comprehensive artifacts. It uses the available workbook honestly, selects a winner from validation evidence, locks the test evaluation, and produces explanations and fairness diagnostics alongside metrics.

Future work should add real policy and document evidence, provider and claimant history windows, verified Indian geography and hospital tiers, optional external boosters, temporal drift evaluation, SHAP/counterfactual audits, and the deep-learning and multi-agent approaches specified in the project prompts. The overall architecture should continue to treat human investigators and policyholders as stakeholders rather than edge cases.

11. Figures and artifact references

The following figures are generated from the run artifacts. They are included to make the report visually inspectable and to provide anchors for a presentation defense.

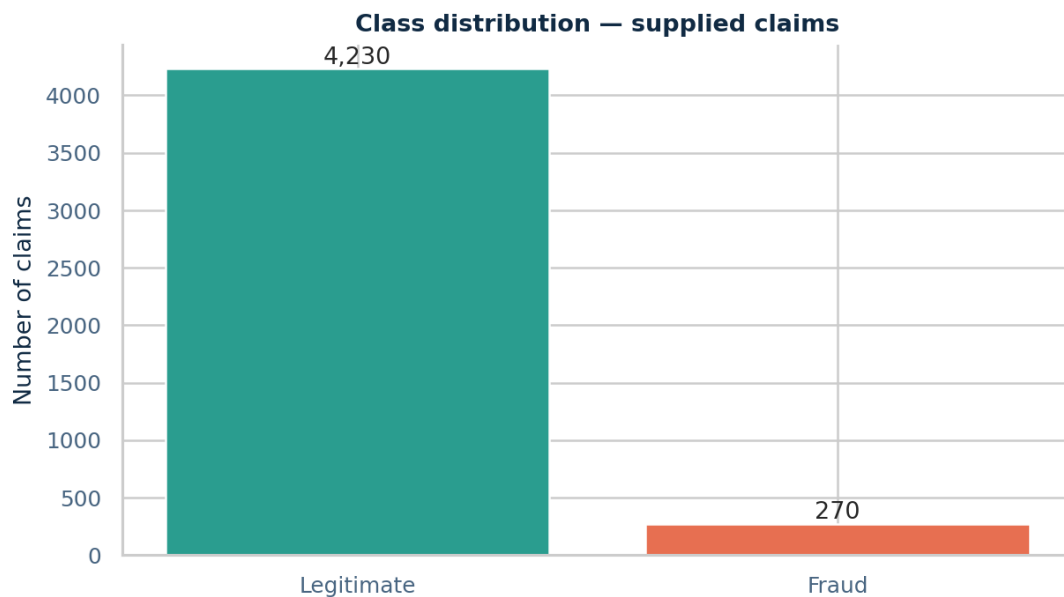


Figure: Class distribution. Fraud is the positive class; bars show all validated rows.



Figure: Fraud prevalence. The supplied snapshot contains a six-percent fraud prevalence.

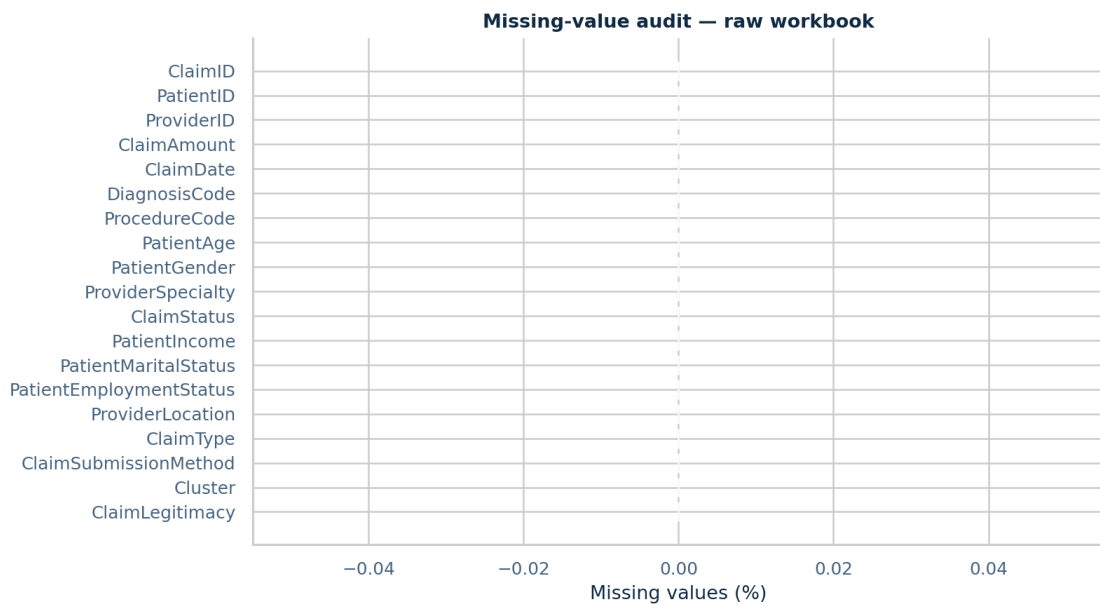


Figure: Missing-value audit. No source column is silently dropped; the pipeline records the measured null rate.

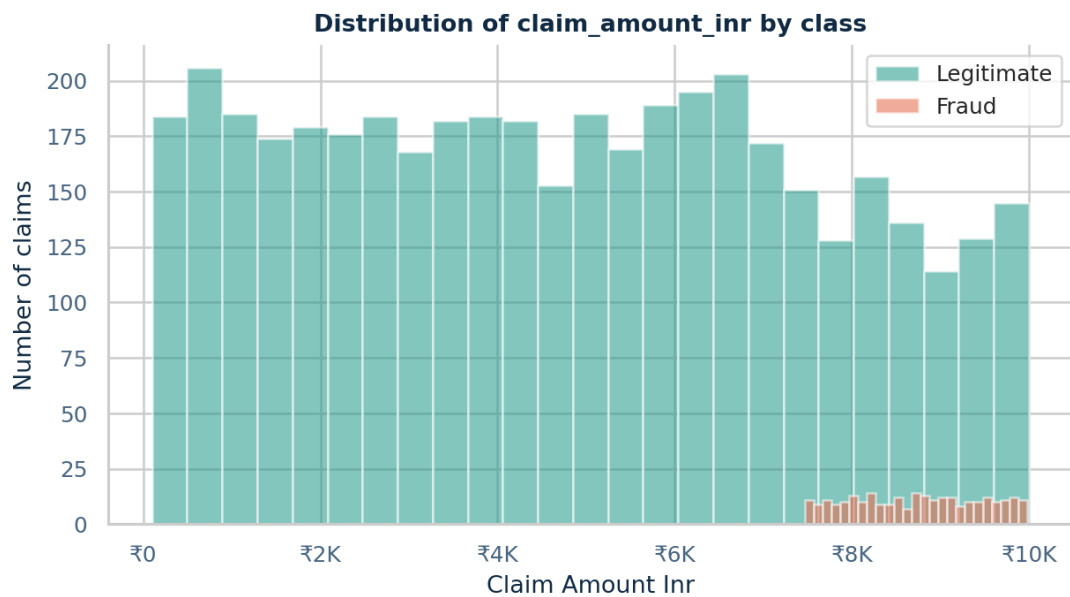


Figure: claim_amount_inr distribution. Training-free distribution view; extreme values are retained for fraud analysis.

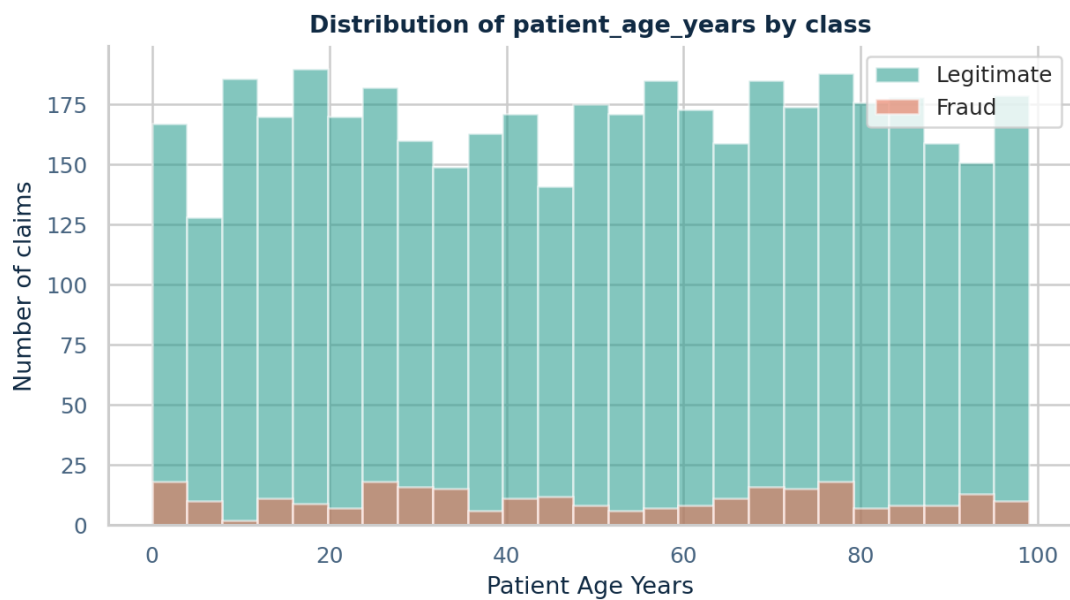


Figure: patient_age_years distribution. Training-free distribution view; extreme values are retained for fraud analysis.

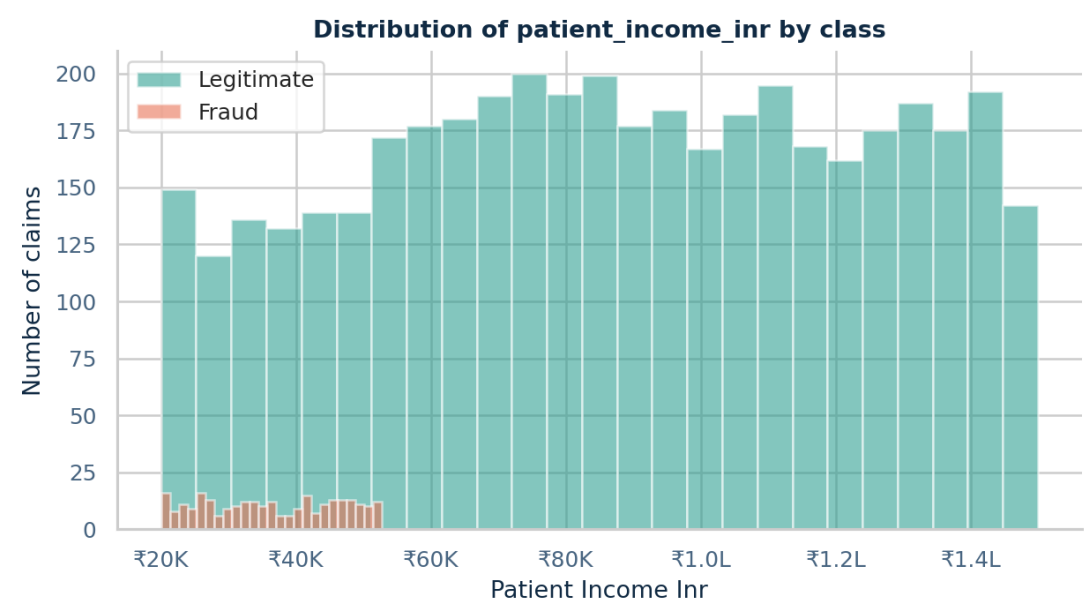


Figure: patient_income_inr distribution. Training-free distribution view; extreme values are retained for fraud analysis.

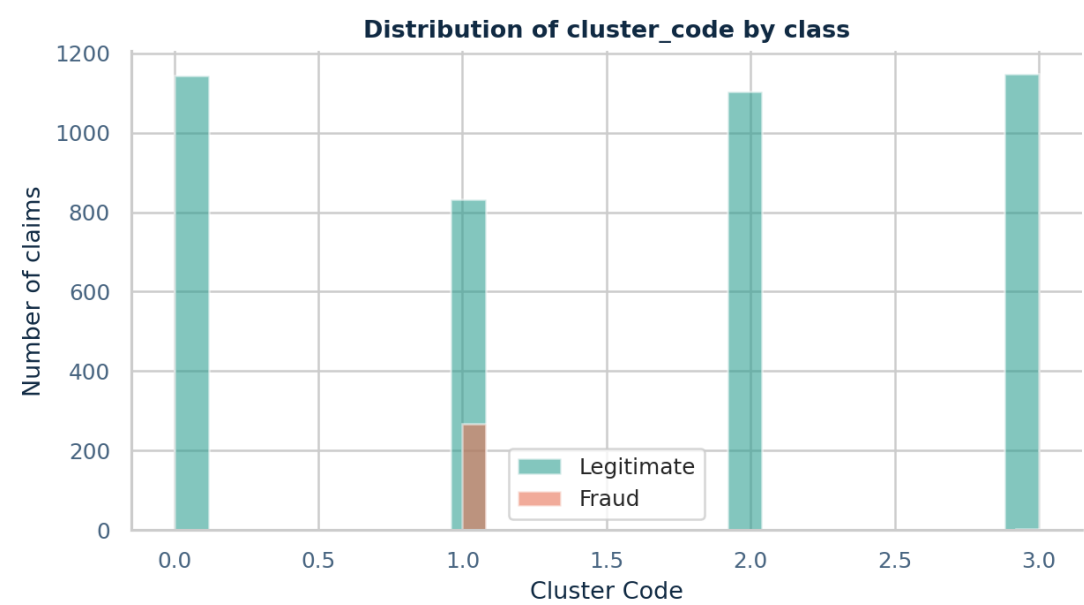


Figure: cluster_code distribution. Training-free distribution view; extreme values are retained for fraud analysis.

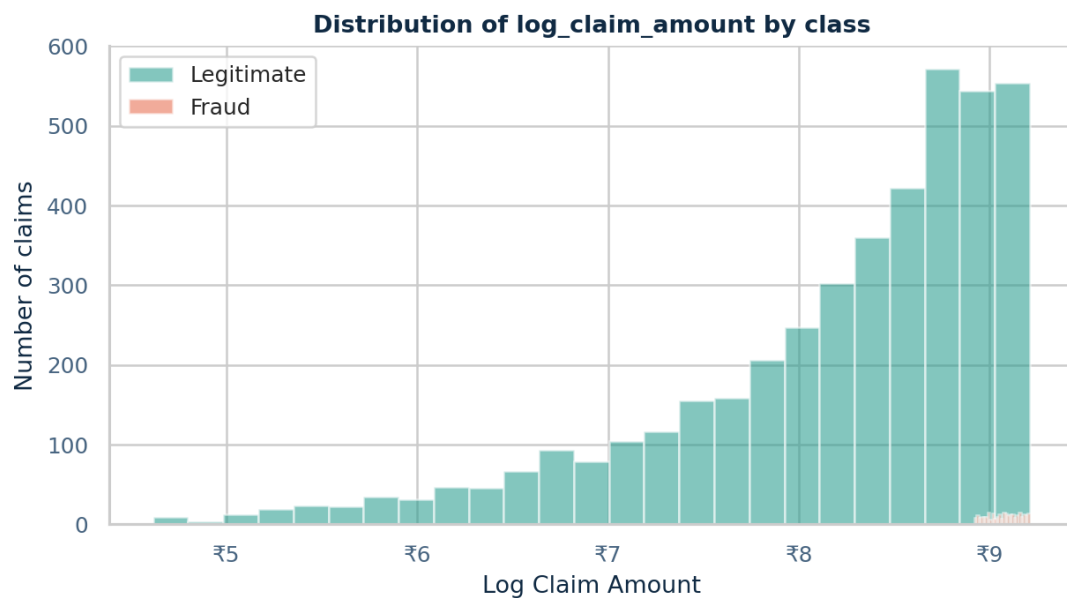


Figure: log_claim_amount distribution. Training-free distribution view; extreme values are retained for fraud analysis.

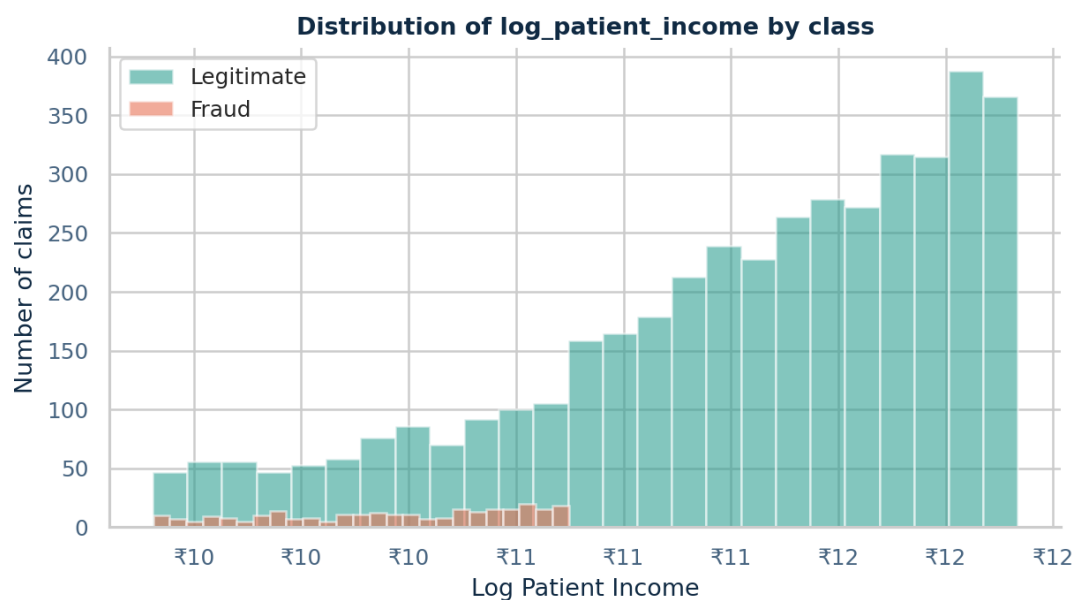


Figure: log_patient_income distribution. Training-free distribution view; extreme values are retained for fraud analysis.

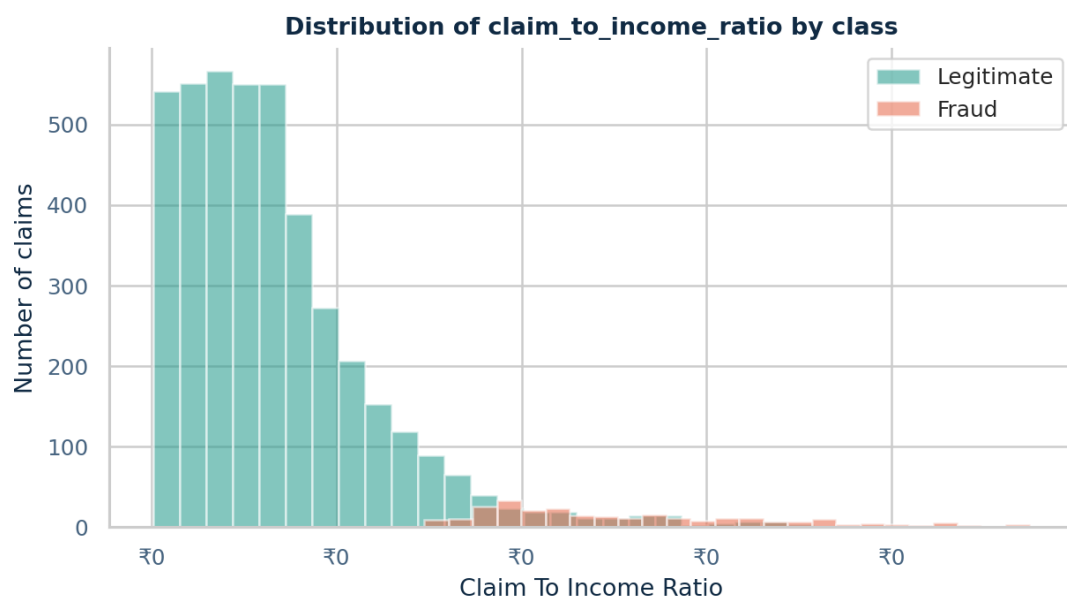


Figure: claim_to_income_ratio distribution. Training-free distribution view; extreme values are retained for fraud analysis.

12. References and acknowledgments

The project acknowledges Prof. Ramesh Athe for academic guidance and IIIT Dharwad for the project context. The complete IEEE-style reference list is maintained in documentation/references.md and reproduced in the companion manuscript.

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23. pandas development team, "pandas documentation," <https://pandas.pydata.org/docs/>.
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25. supplied workbook, `Health Insurance Fraud Claims.xlsx`, repository-provided academic data snapshot; provenance and redistribution license require confirmation from the data owner.

Appendix A — Reproduction and audit checklist

A reviewer should be able to start from the repository, install requirements, execute the single command, inspect the run manifest, open the leaderboard, and trace the report figures back to their source paths. The following checklist is intentionally practical.

Check	Evidence	Status
Input checksum	data/metadata/raw_manifest.json	PASS
Split ratios	evaluation/split_summary.csv	PASS
Feature lineage	documentation/feature_engineering.md	PASS
Model metrics	evaluation/leaderboard.csv and metrics/*.json	PASS
Threshold	evaluation/threshold_memo.md	PASS
Fairness	evaluation/fairness/	PASS
Presentation	presentation/approach_1_traditional_ml.pptx	PASS
PDFs	reports/approach_1_project_report.pdf and approach_1_ieee_paper.pdf	PASS

Appendix B — Complete model inventory

The table below keeps every model row visible, including the baseline. A reviewer can compare the complete suite rather than seeing only the winner.

Each row is generated from the same result dictionary used for evaluation/leaderboard.csv. A blank test value means the row was not the final selected refit, not that a hidden score was omitted.

Rank	Model	Family	Val F2	Val PR-AUC	Test F2
1	Random forest	bagging	1.0000	1.0000	0.9950
2	Gradient boosting	boosting	1.0000	1.0000	0.9799
3	Histogram gradient boosting	boosting	1.0000	1.0000	0.9799
4	AdaBoost	boosting	1.0000	1.0000	0.9799
5	Soft voting ensemble	ensemble	1.0000	1.0000	0.9799
6	Stacking ensemble	ensemble	1.0000	1.0000	0.9950
7	Support vector machine (RBF)	margin	0.9804	0.9820	0.9406
8	Decision tree	tree	0.9804	0.9771	0.9799
9	Extra trees	bagging	0.9709	0.9874	0.9701
10	Logistic regression (L1)	linear	0.9662	0.9775	0.9709
11	Multi-layer perceptron	neural_baseline	0.9615	0.9755	0.9615
12	Linear discriminant analysis	probabilistic	0.9569	0.9133	0.9569
13	Logistic regression (L2)	linear	0.9479	0.9747	0.9615
14	Passive-aggressive classifier	online_linear	0.9479	0.9524	0.9198
15	Calibrated ridge classifier	linear	0.9259	0.9501	0.9242
16	Quadratic discriminant analysi	probabilistic	0.8952	0.7000	0.8547
17	K-nearest neighbors	instance	0.8520	0.8219	0.8028
18	Gaussian naive Bayes	probabilistic	0.8511	0.6629	0.8368
19	Bernoulli naive Bayes	probabilistic	0.8032	0.4821	0.7937
20	Majority-class baseline	baseline	0.2443	0.0607	0.2395

Appendix C — Search-space and budget register

Search spaces are intentionally modest so the single command remains practical on a laptop. This is a reproducible baseline budget, not an exhaustive global hyperparameter optimization.

The `evaluation/tuning/` directory contains one CSV per model. The search score is average precision on stratified training folds; the final threshold is selected later from validation probabilities.

Model	Search	Trials	Parameters
majority	none	1	{}
logistic_l2	grid	3	{"C": 10.0}
logistic_l1	grid	3	{"C": 1.0}
decision_tree	grid	12	{"max_depth": 3, "min_samples_leaf": 3}
random_forest	random	4	{"n_estimators": 220, "min_samples_leaf": 1, "max_depth": null}
extra_trees	random	4	{"n_estimators": 140, "min_samples_leaf": 1, "max_depth": null}
gradient_boosting	random	4	{"n_estimators": 140, "max_depth": 2, "learning_rate": 0.1}
hist_gradient_boosting	random	4	{"max_leaf_nodes": 15, "max_iter": 180, "learning_rate": 0.1, "l2_regu
adaboost	random	4	{"n_estimators": 80, "learning_rate": 0.05}
svm_rbf	random	4	{"gamma": 0.01, "C": 10.0}
knn	grid	6	{"n_neighbors": 5, "weights": "distance"}
gaussian_nb	grid	3	{"var_smoothing": 1e-07}
bernoulli_nb	grid	3	{"alpha": 5.0}
qda	grid	4	{"reg_param": 0.0}
linear_discriminant	grid	2	{"solver": "lsqr"}
mlp	random	4	{"learning_rate_init": 0.003, "hidden_layer_sizes": [64, 32], "alpha":
ridge	grid	3	{"estimator__alpha": 1.0}
passive_aggressive	grid	3	{"estimator__C": 0.1}
voting	none	1	{}
stacking	none	1	{}

Appendix D — Metric dictionary

Accuracy measures all correct rows and is retained for completeness. Precision answers how many flagged rows are fraud. Recall answers how many fraud rows are caught. F1 balances precision and recall; F2 weights recall more heavily.

ROC-AUC and PR-AUC are threshold-free ranking measures. PR-AUC is especially informative when the positive class is rare. MCC and kappa correct for chance agreement. Brier score and log loss examine probability quality. Specificity measures the legitimate-claim true-negative rate.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix E — Threshold and calibration evidence

The operating threshold is frozen from validation F2 before test labels are read. The threshold sweep shows the precision/recall trade-off and is available as a CSV for independent recomputation.

Isotonic calibration is treated as a probability-quality diagnostic. It does not turn a model probability into a legal or clinical conclusion, and it must be recalibrated under distribution shift.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix F — Confusion-matrix interpretation

A true positive is a fraud row flagged by the model; a true negative is a legitimate row left unflagged. A false negative is a fraudulent row that may pass through triage. A false positive is a legitimate row sent to review.

Business review should examine false negatives for missed patterns and false positives for unnecessary burden. The threshold can be changed only through a new validation and governance decision.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix G — Fairness audit protocol

Gender, age band, claim type, and employment slices are scored after the model produces probabilities. They are not direct model identifiers. Small cells are labelled unstable and excluded from strong disparity claims.

A five-percentage-point gap is an investigation trigger. It is not a proof of discrimination because the workbook is a limited supplied snapshot. Mitigation requires better data, legal review, and monitoring.

Slice	Value	N	Fraud N	TPR	FPR	Precision
PatientGender	F	356	25	0.9600	0.0000	1.0000
PatientGender	M	319	15	1.0000	0.0000	1.0000
age_band	0_17	103	5	1.0000	0.0000	1.0000
age_band	18_30	95	7	1.0000	0.0000	1.0000
age_band	31_45	97	8	0.8750	0.0000	1.0000
age_band	46_60	115	4	1.0000	0.0000	1.0000
age_band	61_75	102	5	1.0000	0.0000	1.0000
age_band	76_plus	163	11	1.0000	0.0000	1.0000
ClaimType	Emergency	185	14	1.0000	0.0000	1.0000
ClaimType	Inpatient	159	10	1.0000	0.0000	1.0000
ClaimType	Outpatient	151	6	0.8333	0.0000	1.0000
ClaimType	Routine	180	10	1.0000	0.0000	1.0000
PatientEmploymentStatus	Employed	177	11	1.0000	0.0000	1.0000
PatientEmploymentStatus	Retired	157	15	0.9333	0.0000	1.0000
PatientEmploymentStatus	Student	160	10	1.0000	0.0000	1.0000
PatientEmploymentStatus	Unemployed	181	4	1.0000	0.0000	1.0000

Appendix H — Explainability protocol

Permutation importance shuffles one transformed column on validation rows and measures average-precision change. This supports model debugging and investigator prioritization, not causal inference.

The local explanation template reports score band, strongest signals, evidence values, caveats, and next steps. It explicitly avoids statements about claimant intent and points reviewers to documents and policy terms.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix I — Feature dictionary and exclusions

The complete source dictionary is available as ``data/data_dictionary.csv``. The feature-lineage table is available as ``evaluation/feature_lineage.csv`` and contains the transformed registry used by the estimator.

ClaimID, PatientID, ProviderID, DiagnosisCode, ProcedureCode, ProviderLocation, ClaimStatus, and the target are excluded or transformed for documented reasons. This prevents an impressive-looking score from being driven by memorization.

Feature	Source	Transform
claim_amount_inr	ClaimAmount	renamed
patient_age_years	PatientAge	renamed
patient_income_inr	PatientIncome	renamed
cluster_code	Cluster	numeric category code
log_claim_amount	ClaimAmount	log1p
log_patient_income	PatientIncome	log1p
claim_to_income_ratio	ClaimAmount, PatientIncome	safe ratio
claim_minus_income_scaled	ClaimAmount, PatientIncome	scaled difference
claim_amount_per_age	ClaimAmount, PatientAge	safe ratio
income_age_interaction	PatientIncome, PatientAge	safe ratio
claim_year	ClaimDate	calendar year
claim_month	ClaimDate	calendar month
claim_day_of_week	ClaimDate	weekday index
claim_day_of_month	ClaimDate	day of month
claim_is_weekend	ClaimDate	binary calendar flag
claim_month_sin	ClaimDate	cyclic month encoding
claim_month_cos	ClaimDate	cyclic month encoding
age_band	PatientAge	ordinal bins
cluster_category	Cluster	nominal category
patientgender	PatientGender	normalized category
providerspecialty	ProviderSpecialty	normalized category
patientmaritalstatus	PatientMaritalStatus	normalized category
patientemploymentstatus	PatientEmploymentStatus	normalized category
claimtype	ClaimType	normalized category
claimsubmissionmethod	ClaimSubmissionMethod	normalized category

Appendix J — Data relations and grain

The supplied workbook has a single claim-row grain. Patient and provider identifiers are unique in this snapshot, so historical aggregation cannot be claimed. Future multi-claim data must define policyholder and provider windows before joining.

One-to-many document or history joins must aggregate before returning to one row per claim. Otherwise fan-out can double-count claims and contaminate both training and evaluation.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix K — Reproducibility and environment

The run manifest records the input checksum, configuration, Python environment, model count, feature count, split summary, winner key, and figure count. The serialized preprocessor and selected model provide the inference state.

Reruns should use the pinned requirements file and seed 42. Floating-point or library differences must be recorded rather than silently merged with an earlier run.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix L — Model-card summaries

Every model has a stable key and a metrics JSON file. The model card for the winner includes threshold, validation metrics, test metrics, top features, and the run identifier.

Model cards are not endorsements. They communicate intended use, limitations, data assumptions, and the human-review boundary.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix M — Error-analysis worksheet

The next operational review should group false positives and false negatives by claim amount, age band, claim type, specialty, cluster, and submission method. The current pipeline stores the ingredients for that analysis and avoids printing raw identifiers.

Error analysis should ask whether a misclassification reflects missing policy/document context, a synthetic shortcut, a threshold choice, or ordinary overlap between legitimate and fraud patterns.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix N — Indian-context gap analysis

The implementation uses respectful Indian insurance framing but does not claim that fictional locations are Indian states. Missing fields include policy type, waiting period, co-payment, sum insured, hospital tier, diagnosis descriptions, bills, and IRDAI-facing settlement evidence.

The next data collection should add family-floater and senior-citizen policy cases, cashless and reimbursement modes, public-scheme contexts, provider network status, and multilingual document metadata under a verified license.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix O — Deployment boundary

The code can serialize a model and score a compatible structured row, but cloud deployment, authentication, document ingestion, and live monitoring are outside this traditional baseline.

A production path would need privacy-preserving storage, input validation, model versioning, drift alerts, calibrated thresholds, audit logs, and a human approval workflow.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix P — Future deep and agent approaches

The deep-learning approach should reuse the frozen split, target semantics, metric definitions, and threshold contract. It may add embeddings, learned interactions, and model-specific XAI, but must not move the test goalposts.

The agentic approach should add document verification, policy retrieval, anomaly reasoning, and natural-language explanations. It should combine with this probability as evidence and route disagreements to humans.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix Q — Examiner questions and answers

Why is accuracy not enough? Because the fraud class is only six percent and a majority predictor can appear accurate while catching no fraud. Why is F2 used? It gives recall more weight. Why is the cluster field discussed? Its class concentration may be a synthetic shortcut.

Why is claim status dropped? It may be updated after adjudication. Why use validation before test? To avoid test-set selection bias. Can the model reject claims? No; it is a triage decision-support baseline.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.

Appendix R — Artifact navigation

Start at ``evaluation/run_manifest.json``, open ``evaluation/leaderboard.csv``, inspect the selected model JSON, follow its curve and threshold CSVs, and then open the corresponding figure and document section.

The presentation speaker notes identify the evidence path for each slide. The project report and IEEE paper are generated from the same run context, so a change in numbers requires regeneration rather than manual correction.

Evidence references: ``documentation/complete_technical_manual.md``, ``evaluation/complete_evaluation_record.md``, and the run manifest. These appendices are generated from the same artifact context as the headline results.